

I. LAGRANGIAN FORMULATION

The dynamics of the inverted pendulum are derived using the Euler–Lagrange formulation. The Lagrangian is defined as

$$\mathcal{L} = T - V, \quad (1)$$

where T is the total kinetic energy and V is the gravitational potential energy.

For a system with generalized coordinates q_i , the Euler–Lagrange equations are

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i, \quad (2)$$

where Q_i is the generalized nonconservative force associated with q_i .

A. Coordinate System

The system is described using an inertial Cartesian coordinate frame $\mathcal{F}_I$. The x axis is horizontal and positive to the right, while the z axis is vertical and positive upward.

The position of the wheel axle is

$$\mathbf{r}_w = \begin{bmatrix} x \\ z \end{bmatrix}, \quad (3)$$

where z is constant.

The center of mass of the pendulum body is located at

$$\mathbf{r}_b = \begin{bmatrix} x_b \\ z_b \end{bmatrix} = \begin{bmatrix} x + \ell \sin \theta \\ z + \ell \cos \theta \end{bmatrix}, \quad (4)$$

where ℓ is the distance from the wheel axle to the center of mass of the pendulum body.

The angle θ is measured clockwise from the positive z axis. Therefore,

$$\theta = 0 \quad (5)$$

corresponds to the upright configuration of the inverted pendulum.

The generalized coordinates are

$$q_1 = x, \quad q_2 = \theta, \quad q_3 = \phi_w, \quad (6)$$

where ϕ_w is the angular position of the wheel.

The wheel angular velocity is

$$\omega_w = \dot{\phi}_w. \quad (7)$$

B. Rolling Constraint

Assuming pure rolling without slipping, the wheel translational and rotational velocities satisfy

$$\dot{x} = r_w \dot{\phi}_w, \quad (8)$$

where r_w is the wheel radius.

The corresponding position constraint is

$$x - r_w \phi_w = C, \quad (9)$$

where C is a constant determined by the initial configuration.

Because the rolling constraint eliminates one degree of freedom, ϕ_w may be eliminated from the final equations of motion using

$$\dot{\phi}_w = \frac{\dot{x}}{r_w}, \quad \ddot{\phi}_w = \frac{\ddot{x}}{r_w}. \quad (10)$$

C. Kinetic Energy

We can consider the translational kinetic energy of each of the bodies summed to be equivalent to that of a single body with their combined mass and linear velocity equal to that of the wheel's axis. Since z is constant, and

$$\dot{z} = 0,$$

this lets us derive the translational kinetic energy to be

$$T_{trans} = \frac{1}{2} (m_b + m_w) \dot{x}^2 \quad (11)$$

This approach unfortunately does not account for the translational motion that results from the robot's body. As such we add a correction term for the robot's body translational motion.

$$T_{corr} = \frac{1}{2} m_b \dot{x}^2 m_b \ell \dot{x} \dot{\theta} \cos \theta$$

The rotational kinetic energy of the pendulum body is

$$T_{b,rot} = \frac{1}{2} I_b \dot{\theta}^2, \quad (12)$$

and the rotational kinetic energy of the wheel is

$$T_{w,rot} = \frac{1}{2} I_w \dot{\phi}_w^2. \quad (13)$$

Thus, the total kinetic energy is

$$\begin{aligned} T &= T_{trans} + T_{corr} + T_{b,rot} + T_{w,rot} \\ &= \frac{1}{2} (m_b + m_w) \dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2} (I_b + m_b \ell^2) \dot{\theta}^2 + \frac{1}{2} I_w \dot{\phi}_w^2. \end{aligned} \quad (14)$$

Applying the rolling constraint from (10) gives

$$T = \frac{1}{2} \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta + \frac{1}{2} (I_b + m_b \ell^2) \dot{\theta}^2. \quad (15)$$

D. Potential Energy

The gravitational potential energy of the pendulum body is

$$V = m_b g z_b. \quad (16)$$

Substituting (4) gives

$$V = m_b g (z + \ell \cos \theta). \quad (17)$$

Since z is constant, the term $m_b g z$ is constant and does not affect the equations of motion. It can therefore be omitted, yielding

$$V = m_b g \ell \cos \theta. \quad (18)$$

E. Constrained Lagrangian

Using (15) and (18), the constrained Lagrangian is

$$\begin{aligned} \mathcal{L} &= \frac{1}{2} \left(m_b + m_w + \frac{I_w}{r_w^2} \right) \dot{x}^2 + m_b \ell \dot{x} \dot{\theta} \cos \theta \\ &\quad + \frac{1}{2} (I_b + m_b \ell^2) \dot{\theta}^2 - m_b g \ell \cos \theta. \end{aligned} \quad (19)$$

F. Equations of Motion

The motor applies equal and opposite torques to the wheel and pendulum body. Let M_{mot} denote the motor torque, with positive torque defined in the positive θ direction on the pendulum body.

The generalized motor torque acting on the pendulum is therefore

$$Q_\theta = M_{\text{mot}}. \quad (20)$$

Because the wheel and pendulum body experience equal and opposite motor torques, the corresponding wheel torque is

$$Q_{\phi_w} = -M_{\text{mot}}. \quad (21)$$

After applying the rolling constraint, the motor contributes an effective generalized force in the x direction of

$$Q_x = -\frac{M_{\text{mot}}}{r_w}. \quad (22)$$

The signs in these equations follow the convention that positive θ and positive M_{mot} correspond to clockwise rotation of the pendulum from the upright position.

1) *Translational Equation:* Applying the Euler–Lagrange equation to x gives

$$\left(m_b + m_w + \frac{I_w}{r_w^2}\right)\ddot{x} + m_b\ell \cos\theta\ddot{\theta} - m_b\ell \sin\theta\dot{\theta}^2 = -\frac{M_{\text{mot}}}{r_w}. \quad (23)$$

2) *Pendulum Angular Equation:* Applying the Euler–Lagrange equation to θ gives

$$(I_b + m_b\ell^2)\ddot{\theta} + m_b\ell \cos\theta\ddot{x} - m_b g \ell \sin\theta = M_{\text{mot}}. \quad (24)$$

The first term represents the rotational inertia of the pendulum body, the second represents translational–rotational coupling, and the third represents the gravitational torque.

G. Symbol and Coordinate Definitions

TABLE I
SYMBOLS AND COORDINATE DEFINITIONS.

Symbol	Definition
$\mathcal{F}_I$	Inertial Cartesian coordinate frame.
x	Horizontal position of the wheel axle, positive to the right.
z	Vertical position of the wheel axle, positive upward; treated as constant.
θ	Pendulum-body angle measured clockwise from the positive z axis; $\theta = 0$ corresponds to the upright configuration.
ϕ_w	Angular position of the wheel.
ω_w	Wheel angular velocity, $\omega_w = \dot{\phi}_w$.
$\mathbf{r}_w$	Position vector of the wheel axle.
$\mathbf{r}_b$	Position vector of the pendulum-body center of mass.
x_b, z_b	Horizontal and vertical coordinates of the pendulum-body center of mass.
ℓ	Distance from the wheel axle to the pendulum-body center of mass.
r_w	Wheel radius.
m_b	Mass of the pendulum body.
m_w	Mass of the wheel and axle assembly.
I_b	Mass moment of inertia of the pendulum body about its center of mass.
I_w	Mass moment of inertia of the wheel and axle assembly about the wheel axis.
g	Gravitational acceleration.
T	Total kinetic energy of the system.
V	Gravitational potential energy of the system.
$\mathcal{L}$	Lagrangian, $\mathcal{L} = T - V$.
q_i	i th generalized coordinate.
Q_i	Generalized nonconservative force associated with q_i .
M_{mot}	Motor torque applied between the wheel and pendulum body.
F_f	Horizontal wheel–ground traction force.
C	Constant determined by the initial rolling configuration.
$\mathbf{M}$	Configuration-dependent mass matrix.
$\mathbf{h}$	Velocity-dependent nonlinear term vector.
$\mathbf{g}$	Gravitational generalized-force vector.
$\mathbf{B}$	Motor input vector.
t	Time.